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Novel synthesis of hydroxyapatite via acidic wet chemical precipitation: phase formation, crystallinity and microstructural development

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ABSTRACT

A novel acidic wet precipitation method was developed for the synthesis of hydroxyapatite (HAp), a route largely unexplored due to the propensity for non-apatitic calcium phosphate phase formation under low-pH conditions. This work evaluates the feasibility of this approach and compares it with a conventional alkaline route, focusing on phase evolution, crystallinity, and microstructural development. XRD and Rietveld refinement confirmed that even under acidic conditions HAp remained as the predominant phase after calcination despite its Ca-rich composition (Ca/P ratios between 1.87-1,97). The increase of the calcination temperature from 800 °C to 1000 °C promoted crystal growth and improved structural order, yielding crystallinity values up to 91.26 % and crystallite sizes of 119.58 nm in acidic samples. Complementary thermogravimetric analyses revealed significantly higher volatile content and distinct multistep decomposition profiles for the acidic route, consistent with the nitrate-based precursors and transient phosphate species. TEM images showed denser and less porous morphologies in acidic samples compared to their alkaline counterparts. The results demonstrate that acidic precipitation is a viable pathway to obtain crystalline HAp when synthesis and calcination conditions are precisely controlled, offering an alternative route for future dopant incorporation and advance bioceramic powder design.

1. Introduction

Calcium phosphate (CaP) bioceramics are extensively applied in biomedical fields due to their compositional similarity to the mineral phases of bone tissue and teeth. Among them, hydroxyapatite [HAp, $Ca_{10}(PO_4)_6(OH)_2$] is the most investigated, exhibiting excellent biocompatibility, bioactivity and osteoinductivity [1]. Synthetic HAp is widely used in scaffolds for tissue engineering [2–4], bioactive coatings [5–8], bone fillers, and drug delivery systems [9,10]. Various methods are available to produce synthetic hydroxyapatite, which can be categorized into dry, wet, high-temperature and biogenic methods. The synthesis route strongly influences HAp's morphology, crystallinity, and phase purity, which in turn affect biological and mechanical performance [11]. Wet chemical precipitation remains a versatile method due to its simplicity, cost-effectiveness and scalability, but the resulting material properties depend critically on pH, temperature, precursor type, and reaction kinetics.

One of the most relevant properties of HAp is its structural capacity to absorb anions and cations, enabling ionic exchange within hard tissues [12,13]. Both cationic (e.g., Sr^{2+} [14–18], Mg^{2+} [19–21], Zn^{2+} [15,22,23], and Al^{3+} [24,25]), and anionic (e.g., Cl^- [26,27], and F^- [28–30]) substitutions can significantly tailor solubility, mechanical strength, and biological response [12,31]. Recent studies have also explored dual ionic substitutions in HAp, such as Mg^{2+}/La^{3+} [32], Mg^{2+}/Zn^{2+} [33], or Cu^{2+}/Zn^{2+} [34], the latter aimed at enhancing the electrical and biological properties of the materials.

Recently, niobium (Nb) has gained considerable attention as a dopant for HAp, due to its well-documented biocompatibility as a coating material and alloy component for biomedical applications [35]. The incorporation of Nb ions to the HAp's lattice achieved good biocompatibility, osteoinductivity and bioactivity, and promoted antibacterial properties to the materials [13,36–39]. Thiyagarajan et al.

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[13], studied Nb substitution through density functional theory (DFT), postulating that different incorporation mechanisms would take place in acidic and alkaline conditions, as experimentally observed by Capanema et al. [37] and Tamai et al. [36]. This effect would allow control over the lattice incorporation pathway simply by the change of the precipitation medium's pH. However, acidic wet synthesis routes for HAp have remained unexplored, as low pH conditions typically favor the precipitation of non-apatitic calcium phosphate phases such as brushite ($CaHPO_4$. $2H_2O$) and monetite ($CaHPO_4$).

Nevertheless, Thürmer et al. [40] demonstrated that acidic aqueous reactions can yield tricalcium phosphates (TCPs) without non-apatitic CaP formation. Since most TCP syntheses through wet chemical reactions pass through a calcium-deficient hydroxyapatite (CDHA) intermediate before calcination [26], this could suggest that direct HAp precipitation under acidic conditions may be feasible if the synthesis is properly controlled. Although the proposition of an acidic route is not primarily intended for improvement of the synthesis compatibility with large-scale industrial production, addressing this gap could significantly expand the current current understanding of HAp precipitation mechanisms and improve the design window for HAp-based ceramics, as it provides a controlled platform to investigate the role of pH on phase stability, crystal growth, and ion incorporation mechanisms in hydroxyapatite.

The present study proposes, hence, a new acidic precipitation route for HAp synthesis systematically comparing it with a conventional alkaline process. Comprehensive structural, chemical, and morphological analyses were employed to elucidate phase evolution, thermal behavior, and microstructural development, in order to provide new insights into the stability of HAp synthesis under acidic conditions and its implications for the design of next-generation bioactive ceramics.

2. Materials and methods

2.1. Alkaline synthesis

$$10Ca(OH)_2 + 6H_3PO_4 \rightarrow Ca_{10}(PO_4)_6(OH)_2 + 18H_2O \quad (1)$$

HAp powders were synthesized through a conventional alkaline wet chemical precipitation route. A 0.5M orthophosphoric acid solution [H_3PO_4, Dinâmica, Brazil] was added dropwise to a 0.5M calcium hydroxide suspension [$Ca(OH)_2$, Dinâmica, Brazil] under continuous stirring at 300 rpm. The molar ratio between Ca^{2+} and PO_4^{3-} ions (Ca/P ratio) was fixed at 1.67 to obtain stoichiometric HAp, following the reaction described in Eq. (1). The pH of the suspension was monitored using a calibrated pH meter (Onesense PH2500, Brazil) and stabilized at approximately 13, confirming the strongly alkaline conditions favorable for HAp precipitation. The suspension was maturated at 90 °C for 24 h, then vacuum-filtered through a Büchner funnel and rinsed three times with 95 % ethanol.

The resulting precipitate was dried in a furnace at 100 °C for 24 h and calcined at either 800 °C or 1000 °C for 1 h. The calcined powders was manually ground with a pestle and mortar and sieved through a #325 mesh (44 μm) sieve for further use. Samples prepared by this route were designated as HApK800 and HApK1000, indicating the synthesis route (alkaline, "K") and calcination temperature of 800 °C or 1000 °C, respectively.

2.2. Acidic synthesis

$$10Ca(NO_3)_2 \cdot 4H_2O + 6H_3PO_4 \rightarrow Ca_{10}(PO_4)_6(OH)_2 + 20HNO_3 + 38H_2O \quad (2)$$

A novel acidic precipitation route was developed using calcium nitrate tetrahydrate [$Ca(NO_3)_2{\cdot}4H_2O$, Êxodo, Brazil] as the calcium source to maintain acidic conditions through the formation of nitric acid [HNO_3], as described in Eq. (2). The synthesis was performed by the dropwise addition of a 0.5M H_3PO_4 solution to a 0.5M $Ca(NO_3)_2{\cdot}4H_2O$ solution, keeping the Ca/P molar ratio constant at 1.67.

Unlike the alkaline synthesis, no immediate precipitation occurred due to the low pH (~0.5). The solution was maintained at room temperature for 5min, then subjected to solvent evaporation on a hot plate under continuous magnetic stirring for 5 h, maintaining a solution temperature of approximately 125 °C and allowing complete evaporation of the solution. The pH was constantly assessed to guarantee that the pH remained around 0.5.

The resulting residue was calcined at either 800 °C or 1000 °C for 1 h, then ground and sieved as previously described in section 2.1. These samples were denoted as HApA800 and HApA1000, referring to the acidic ("A") synthesis route and calcination temperatures.

2.3. Phase composition and crystallography

The identification of crystalline phases in the synthesized powders was carried out using X-ray diffraction (XRD) with an Aeris diffractometer (PANalytical, The Netherlands) employing CuKα radiation source (λ = 1.5406 Å). The measurement spanned from 20° to 70° (2θ) with a step size of 0.01° and an integration time of 50s per step.

Quantitative phase analysis and lattice parameter determination were performed via Rietveld refinement using Profex 5.6.0 software and employing the structures standards ICSD 026204 (HAp), ICSD 006191 (β-TCP) and ICSD 014922 (CaO). Additionally, the crystallite size was estimated from the (211) diffraction plane of HAp, as obtained from the Rietveld refinement performed in Profex software. The crystallinity index (%C) was calculated according to Eq. (3):

$$\%C = \left(1 - \frac{V_{112/300}}{I_{300}}\right) \times 100 \quad (3)$$

Where $V_{112/300}$ is the intensity of the valley between the (112) and (300) peaks, and I_{300} corresponds to the maximum intensity of the peak associated with the (300) crystallographic plane [41].

The elemental composition of the powders was investigated by X-ray fluorescence (XRF) using a Rigaku X-ray RIX 2000 spectrometer equipped with a rhodium (Rh) X-ray tube, primarily to determine the Ca/P ratio and to assess the presence of any additional elements in the powders. The samples were prepared as wax-pressed pellets from the dried powders.

Fourier transform infrared spectroscopy (FTIR) was performed on a PerkinElmer IR spectrometer equipped with an ATR accessory, covering the range of 450 cm^{-1} to 4000 cm^{-1}, to confirm the presence of the HAp's chemical groups and structures. This analysis was applied as a complementary assay to XRD and XRF results.

2.4. Thermal evolution of obtained powders

The thermal evolution of alkaline and acidic powders was assessed through thermogravimetric analysis (TGA) using a TGA4000 (PerkinElmer, USA) system. These assessments were performed to verify possible differences in the thermal behavior and development of the obtained calcium phosphates, and, if needed, to posteriorly assist in their phase identification. Measurements were performed under nitrogen atmosphere from 30 °C to 950 °C at a heating rate of 10 °C/min, using Al_2O_3 crucibles.

2.5. Powder morphology and nanostructure

The morphology and nanostructure were examined by transmission electron microscopy (TEM) using a JEOL 1400FLASH (120 kV) microscope. Particle shape, agglomeration state, and porosity were analyzed to correlate morphological features with synthesis conditions and thermal treatments.

2.6. Statistical analysis

All synthesis experiments were performed in triplicate (n = 3). Rietveld refinement, crystallite size, and crystallinity calculations were conducted independently for each replicate. Results are reported as mean ± standard deviation (SD). Statistical significance was evaluated by one-way ANOVA followed by Tukey's multiple comparison test, with $p < 0.05$ considered significant.

3. Results and discussion

3.1. Phase composition and structural analysis

XRD patterns of all synthesized powders (Fig. 1) exhibited the characteristic peaks of HAp (ICSD 026204) at 2θ = 22.8°, 25.9°, 31.7° and 39.7°. The Rietveld refinement (Table 1) confirmed HAp as the predominant phase in all samples, with minor secondary phases depending on the synthesis medium and calcination temperature.

In the alkaline route, both HApK800 and HApK1000 displayed high phase purity (>98 %), with only trace β-tricalcium phosphate (β-TCP) detected in HApK800. In contrast, the acidic route led to the identification of calcium oxide (CaO) peaks, particularly at higher calcination temperatures (HApA1000), indicating partial thermal decomposition of HAp. The presence of CaO is particularly relevant from a biomedical perspective: when in contact with aqueous or physiological media, CaO reacts to form calcium hydroxide, which increases the local pH and may cause cytotoxic effects or compromise cell adhesion during the early stages of implantation. Additionally, CaO formation can impair coating adhesion and mechanical integrity when HAp is used as a bioactive layer [42,43].

The existence of these secondary phases could be due to two main factors: chemical and thermodynamic mechanisms. According to Demnati [44], the formation of coexistent phases with HAp between temperatures of 850 °C and 1000 °C is dependent exclusively on the apatite's Ca/P molar ratio. Stoichiometric hydroxyapatite has a Ca/P ratio of 1.67, which was adopted during the two different synthesis mechanisms performed in this work. However, in acidic media, phosphate groups can present a different degree of dissociation and higher protonation, forming H_3PO_4 or $H_2PO_4^-$, which would alter the availability of phosphate groups for the synthesis of HAp. Consequently, this could cause an increase in the Ca/P ratio of the synthesized HAp,

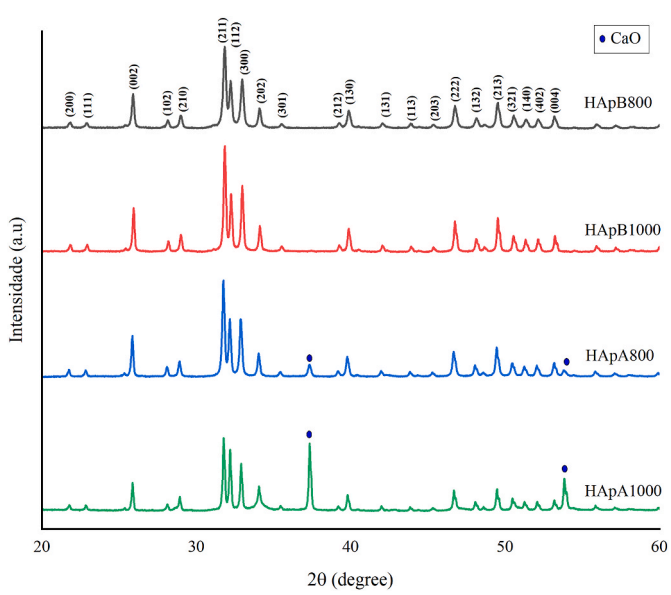

Fig. 1. X- Ray diffractograms of HApK800, HApK1000, HApA800 and HApA1000 samples.

favoring the formation of Ca-rich secondary phases during calcination [45]. Consistent with this interpretation, the identification of CaO on samples prepared in acidic medium reflects an excess of calcium not incorporated into the apatite lattice, supporting the notion that the acidic synthesis conditions biased the system toward Ca-rich compositions due to phosphate availability. Furthermore, careful inspection of the diffractograms revealed no diffraction peaks corresponding to tetracalcium phosphate (TTCP, $Ca_4(PO_4)_2O$), indicating that this phase was not formed under the synthesis and heat-treatment conditions employed in the present work.

Despite the variation in phase composition, the lattice parameters and unit cell volume remained nearly constant across all samples (a≈9.42 Å, c ~ 6.88 Å), confirming that the hexagonal HAp structure was retained even under acidic synthesis conditions.

The refinement reliability parameters (Rwp, Rexp, GoF and χ^2) were used to evaluate the quality and reliability of the structural refinements for the samples (Table 2). The Rwp values range between 7 and 8 %, with Rexp values around 6–7 %, resulting in GoF factors close to unity (1.1–1.2). These results indicate that the refinements were of good quality and that the calculated patterns reproduced well the experimental data. The χ^2 values around 1.3–1.5 also support the reliability of the obtained structural models [46].

Crystallinity and crystallite size of the HAp samples are summarized in Table 3. Both parameters increased with calcination temperature independently of the synthesis route, confirming that thermal energy enhanced lattice ordering and grain growth. The acidic samples exhibited a stronger temperature response, with HApA1000 achieving the highest crystallinity (91.26 %) and crystallite size (119.6 nm). Conversely, HApK800 showed the lowest values (82.26 % and 53.22 nm). These results suggest that acidic conditions delay nucleation but promote the growth of larger, more ordered crystallites once precipitation occurs.

The influence of acidic conditions on hydroxyapatite crystallization remains a matter of debate in literature. The obtained results would be corroborated by the predominant view that low pH decreases nucleation rates, leading to fewer but larger crystallites with higher structural order – supported by reports by Zhang and Darvell [47], who reported that low pH and high Ca/P conditions reduce the rate of supersaturation and nucleation, favoring the formation of larger hydroxyapatite. This perspective is further corroborated by subsequent investigations into calcium orthophosphate systems. For instance, in the context of phosphorus recovery from alkaline phosphate concentrates, lower pH conditions (e.g., pH 8 compared to 11.5) coupled with stoichiometric Ca/P ratios facilitate direct hydroxyapatite nucleation without metastable precursors, leading to reduced nucleation stages and the formation of larger crystallites with higher degrees of crystallinity [48]. Similarly, recent syntheses of high-aspect-ratio nanostructured hydroxyapatite demonstrate that acidic environments enhance anisotropic crystal growth along the c-axis, attributed to lowered supersaturation and nucleation rates, resulting in elongated nanostructures with improved aspect ratios [49].

Statistical analysis (one-way ANOVA) indicated a specific significant difference between the groups for both crystallinity and crystallite size ($p < 0.0001$). Regarding crystallite size, Tukey's post hoc test showed that a sample synthesized in an alkaline medium at 800 °C (HApK800) differed significantly from the other samples. HApK1000 and HApA800, on the other hand, showed no significant differences between themselves ($p = 0.313$), while HApA1000 stood out with the highest values, differing from all others. In the crystallinity analysis, comparisons between the groups revealed that HApK1000 and HApA800 differed significantly from HApK800 ($p < 0.001$), but not from each other. HApA1000 exhibited a significant difference only relative to HApK800 ($p < 0.0001$) and showed no significant difference from HApK1000 ($p = 0.556$) or HApA800 ($p = 0.170$). Taken together, these emphasize the importance of both synthesis route and thermal treatment in shaping the final materials properties and indicate that acidic precipitation followed

Table 1
Rietveld refinement and unit cell parameters.

Sample	Phase percentage (%)			Lattice parameters (Å)		
	HAp	β-TCP	CaO	*a*	*c*	U.C.V
HApK800	98.49 ± 1.5	1.5 ± 1.5	0.01 ± 0.01	9.41	6.88	529.38
HApK1000	99.27 ± 0.4	0.52 ± 0.26	0.23 ± 0.11	9.42	6.88	530.77
HApA800	91.41 ± 6.22	0.87 ± 1.24	7.71 ± 6.99	9.42	6.88	530.28
HApA1000	85.27 ± 14.68	0.93 ± 0.42	13.79 ± 14.93	9.42	6.88	529.70

Values are expressed as mean ± standard deviation (n = 3).

Table 2
Rietveld refinement parameters of hydroxyapatite powders.

Sample	Rwp	Rexp	GoF	χ^2
HApK800	7.26 ± 0.49	6.46 ± 0.04	1.12 ± 0.08	1.27 ± 0.18
HApK1000	7.93 ± 0.10	6.60 ± 0.02	1.20 ± 0.01	1.44 ± 0.04
HApA800	7.90 ± 0.40	6.48 ± 0.27	1.21 ± 0.01	1.48 ± 0.03
HApA1000	7.93 ± 0.59	6.66 ± 0.03	1.19 ± 0.08	1.42 ± 0.20

Values are expressed as mean ± standard deviation (n = 3).

Table 3
Crystallinity and crystallite size of the samples.

Sample	Crystallinity (%)	Crystallite size (nm)
HApK800	82.26 ± 1.52	53.22 ± 5.18
HApK1000	89.89 ± 0.09	94.37 ± 3.96
HApA800	88.91 ± 0.76	81.05 ± 2.02
HApA1000	91.26 ± 1.78	119.58 ± 0.94

Values are expressed as mean ± standard deviation (n = 3).

by high-temperature calcination is the most effective combination for crystallite growth.

As summarized in Table 4, the acidic wet precipitation route developed in this work produced hydroxyapatite with crystallinity between 88.91 and 91.26 % and crystallite size in the range of 81.05–119.58 nm after calcination at 800–1000 °C. these values are comparable to, and in some cases higher than, those reported for hydrothermal, sol-gel, and conventional wet precipitation routes. The results indicate that is highly effective in producing structurally ordered hydroxyapatite and represents a robust platform for investigating pH- dependent phase stability and ion incorporation.

XRF analysis revealed distinct Ca/P ratios between the alkaline and acidic routes (Table 5). The alkaline powders were near stoichiometric (Ca/P = 1,78-1.87), whereas the acidic samples (HApA800 and HApA1000) exhibited elevated Ca/P molar ratios (2.06 and 2.15, respectively). These findings suggest that, under acidic conditions, the limited incorporation of phosphate into the lattice promotes calcium excess, which segregates as CaO upon calcination. In addition, phosphate volatilization during high-temperature treatment may further

Table 4
Comparison of crystallinity and crystallite size of hydroxyapatite synthesized by different methods reported in the literature and by the acidic wet precipitation route developed in this work.

Synthesis Method	Calcination Temperature (°C)	Crystallinity (%)	Crystallite size (nm)	Reference
Hydrothermal	–	79–99	15–17	[50]
	–	55.83–86.13	28.76–89	[51]
Sol-gel	800	88	61	[52]
	700	84	95	[53]
Wet precipitation	700	50–64	90–100	[54]
	650	72	47.4	[55]
Wet precipitation acidic route	800	88.91	81.05	This work
	1000	91.26	119.58	

Table 5
Ca/P molar ratio determined by X-ray fluorescence (XRF).

Sample	Ca/P molar ratio
HApK800	1.78
HApK1000	1.87
HApA800	2.06
HApA1000	2.15

Table 6
Loss on ignition (LOI) and trace oxides detected by X-ray fluorescence (XRF) – values in wt%.

Sample	Impurities				LOI
	Mg	Na	Mn	Si	
HApK800	0	0	0.005	0	1.5
HApK1000	0	0.043	0.006	0	1.0
HApA800	1.450	0.007	0.001	0	4.3
HApA1000	1.525	0	0.001	0	3.4

contribute to the Ca enrichment, as reported in similar calcium phosphate systems [56,57]. Despite these effects, the HAp phase remains dominant, and the lattice parameters and unit cell volumes were comparable to those of the alkaline samples, indicating that the core HAp structure was preserved.

Moreover, XRF showed minor constituents and impurities in the samples, such as Mg, Na and Mn, which were detected in trace amounts (Table 6).The presence of Mg, mainly observed the acidic samples, may be originated from the $Ca(NO_3)_2{\cdot}4H_2O$ precursor [58,59]. Commercial calcium nitrate tetrahydrate, typically produced from natural sources such as limestone or dolomite, often contains trace amounts of Mg incorporated in its mineral matrix, which persists despite industrial purification. In contrast, the basic synthesis route employs purer calcium hydroxide, derived from refined calcium oxide precursor, minimizing such impurities [60]. Sodium and manganese oxides were detected in lower concentrations and are likely derived from minor impurities in the reagents or cross-contamination during sample preparation.

In addition to elemental composition, the Loss on Ignition (LOI) was also evaluated. LOI is a gravimetric parameter that quantifies the percentage of mass lost during heating of the samples to high temperatures, usually due to the release of volatile compounds such as structural water, carbonates, nitrates, or other thermally label species. In the present study, LOI values ranged from 1.0 % to 4.3 %. Higher LOI values were observed in the acidic synthesis route, which may be related to the greater reactivity of these samples, favoring adsorption of atmospheric moisture, presence of residual nitrates or water molecules trapped in the structure.

FTIR spectra, presented in Fig. 2, reveal the chemical groups within the samples. In all samples, characteristic functional groups of HAp were observed. The phosphate group (PO_4^{3-}) bands in the 570-600 cm^{-1} region along with the peak at 630 cm^{-1} associated with the vibration of OH^- ions creates the characteristic HAp triplet. Moreover, another peak related to the OH^- group is observed at 3570 cm^{-1}.

Tetrahedral ions or molecules, such as phosphate, can exhibit four

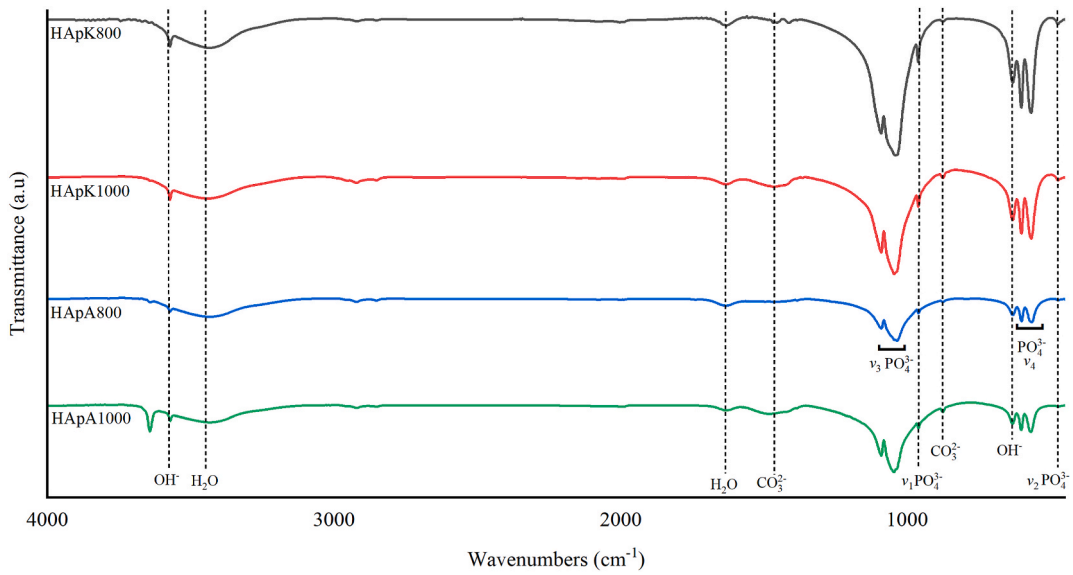

Fig. 2. Fourier transform infrared spectra of the samples.

types of vibrational modes: ν_1, ν_2, ν_3, ν_4 [45]. The bands at 1030 cm^{-1} and 1090 cm^{-1} are attributed to the ν_3 stretching modes of phosphates, while the band at 960 cm^{-1} corresponds to the ν_1 symmetric stretching mode. Additionally, a less intense phosphate group peak is observed at 470 cm^{-1}, associated with the ν_2 bending vibration mode of phosphate. Bands in the 3700-3000 cm^{-1} range and around 1630 cm^{-1} correspond to adsorbed water.

Based on the FTIR spectra, the presence of carbonate groups (CO_3^{2-}) was also identified in the 1450 cm^{-1} region, indicating the substitution to some degree of phosphate groups by CO_3^{2-} ions in the hydroxyapatite structure. This substitution is expected due to the absorption of atmospheric CO_2 during the synthesis process (solution preparation, filtration, drying, and calcination) carried out under air atmosphere [45]. In addition to the atmosphere, the calcination temperature also plays a relevant role in this incorporation. Higher temperatures can increase the reactivity and diffusion of CO_2 into the structure, promoting the formation of carbonate-substituted hydroxyapatite, as reported in other studies [61,62]. Furthermore, in the HApA1000 sample, a band at 3643 cm^{-1} is observed, which can be attributed to the stretching vibration of structural hydroxyl groups associated with the presence of CaO, in agreement with the XRD results, confirming the presence of this material [63,64].

3.2. Thermal evolution of precursor powders

The results of the thermal analyses of the obtained hydroxyapatite precursors are shown in Fig. 3, revealing markedly different thermal profiles for the different synthesis routes.

The sample obtained in alkaline medium exhibited a smooth and gradual weight loss of approximately 4 % up to 950 °C, mainly attributed to the release of physically adsorbed water and structural dehydroxylation of hydroxyapatite. The first thermal event (observed in both alkaline and acidic samples) occurs in the range of 50° to 65 °C and is related to the loss of adsorbed water on the surface of the particles. Precipitated HAp contains two types of water: physically adsorbed, which can be removed at low temperatures (25°–200 °C), and chemically adsorbed, which is structurally bound and requires more energy to remove, only occurring in the range of 200°–500 °C [65]. With the increase in temperature (600°–800 °C), there is also a mass loss associated with the dehydroxylation of hydroxyapatite, resulting in the removal of OH- ions from the structure [42,55].

In contrast, the sample synthesized in acidic medium showed a

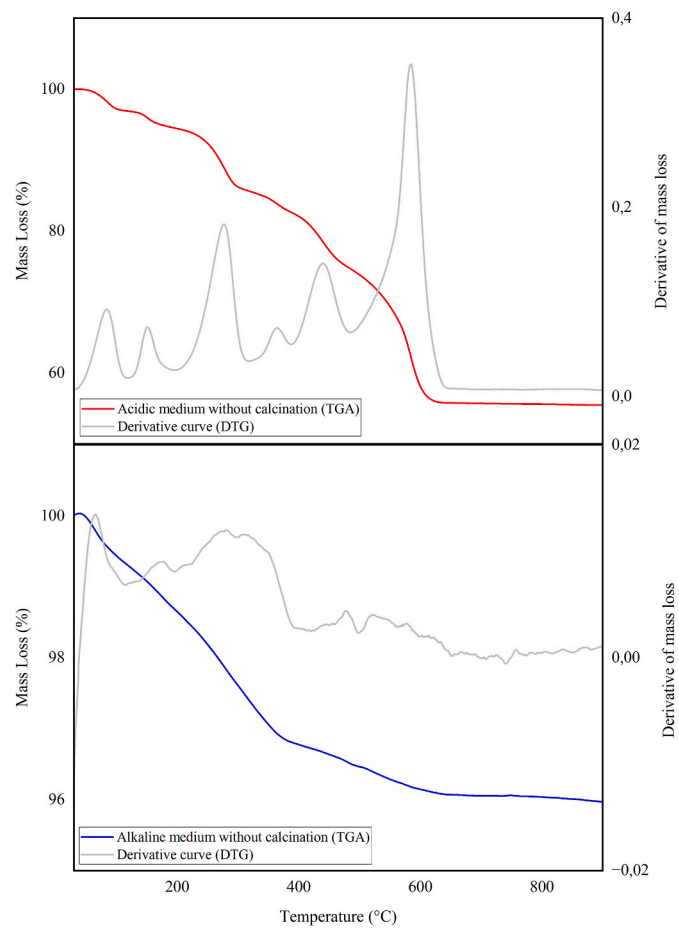

Fig. 3. Thermogravimetric analysis (TGA) and derivative thermogravimetry (DTG) curves of samples before calcination.

significant mass loss (~45 %) with multiple distinct decomposition steps up to around 600 °C. The pronounced mass loss observed in the acidic sample can be attributed primarily to the calcium nitrate tetrahydrate precursor ($Ca(NO_3)_2{\cdot}4H_2O$), which contains a high amount of crystal water. As reported in the literature [66], the thermal decomposition of

this salt proceeds in successive steps of dehydration followed by nitrate breakdown, producing CaO as the stable residue. This staged process accounts for the mass losses with an onset detected around 65 °C and 230 °C, as well as the more significant degradation with an onset between 550 and 600 °C. In addition to this contribution, the powder also contains dicalcium phosphate dihydrate (DCPD, CaHPO4·2H2O), a phase likely formed due to the presence of unreacted calcium nitrate and the comparable solubility constants of calcium phosphates and hydroxyapatite in strongly acidic media [45,67]. The presence of DCPD introduces further weight loss events, beginning with dehydration with an onset at approximately 140 °C to form anhydrous dicalcium phosphate (DCPA, CaHPO4). With increasing temperature, DCPA undergoes gradual conversion into calcium pyrophosphate (CPP, Ca2P2O7) producing additional losses with onset at 340–405 °C range [63,64]. Finally, the major decomposition of calcium nitrate, with an onset near 550 °C coincides with the initiation of the solid-state reaction between residual CaO and CPP, marking the last significant step in the overall thermal sequence. This multistep profile aligns with the higher LOI values observed for the acidic samples (3.4–4.3 %).

These results demonstrate that the acidic synthesis route produces powders with higher volatile content and lower initial thermal stability, which directly influences the final composition and crystallinity of the calcined materials.

3.3. Morpholy and nanostructure

Some factors, such as synthesis method and the specific characteristics of each process, are essential in determining the morphology of the particles [55,65]. Transmission electron microscopy (TEM) micrographs revealed clear morphological differences among hydroxyapatite samples (Fig. 4), directly influenced by the synthesis medium and calcination temperature, and in agreement with the findings from XRD, XRF, FTIR and TGA analyses. The samples synthesized in alkaline medium and calcined at 800 °C (a) and 1000 °C (b) exhibited predominantly rod-like nanoparticles with well-defined contours and visible interparticle porosity, especially in sample (a). Although porosity may compromise mechanical performance, it significantly enhances the biological compatibility of the material, as it facilitates fluid exchange, protein adsorption and cell adhesion [68]. The increase in calcination temperature to 1000 °C led to grain growth, resulting in larger particles with some faceted edges and a slight reduction in porosity. According to Figueiredo et al. [69], calcination temperature has a significant impact on the microstructure of hydroxyapatite. While crystallinity and particle size increases with temperature, porosity decreases strongly at elevated temperatures.

In contrast, the samples synthesized in acidic medium (c and d) presented denser and more irregular agglomerates without evident porosity. Sample (c), calcined at 800 °C, showed poorly defined and compact structures, which may be related to its high Ca/P molar ratio and the consequent formation of secondary phases. As reported by Wang et al. [70], even slight derivations from the stoichiometric Ca/P ratio of 1.67 can lead to the appearance of calcium oxide (CaO) or tricalcium phosphate (TCP), both of which affect the microstructural stability and degradation behavior of hydroxyapatite. In particular, the presence of CaO, even in small amounts, has been shown to promote localized grain boundary weakening and trigger the formation of new surface phases, especially under physiological or mildly acidic conditions. Although the sample calcined at 1000 °C (d) exhibited slightly more defined grains, the morphology remained compact and less organized, lacking the porosity observed in the basic samples. The results confirm that the synthesis medium plays a critical role in determining the final morphology of hydroxyapatite and the choice of calcium precursor can also significantly influence the structural characteristics of the resulting material. Previous studies have demonstrated that the selection of calcium precursors affects not only the crystallinity and thermal behavior, but also the morphology and agglomeration state of hydroxyapatite powders [71,72].

4. Conclusion

This work demonstrated that phase-pure and highly crystalline hydroxyapatite can be successfully synthesized via an acidic wet chemical precipitation route, despite the unfavorable conditions traditionally associated with HAp formation. X-ray diffraction and Rietveld refinement confirmed HAp as the predominant phase after calcination, with only minor formation of calcium oxide (CaO) at higher temperatures. Increasing the calcination temperature from 800 °C to 1000 °C promoted crystal growth and structural ordering, with the HApA1000 sample reaching a crystallinity of 91.26 % and a crystallite size of 119.58 nm.

The hexagonal crystal structure was retained for all samples, with consistent lattice parameters (a ≈9.42 Å and c ≈ 6.88 Å). XRF analyses showed an increase in the Ca/P molar ratio in acidic samples, consistent with phosphate limitation and the formation of calcium-rich secondary phases. Thermogravimetric analyses revealed higher volatile content and distinct dehydration–decomposition behavior for acidic precursors.

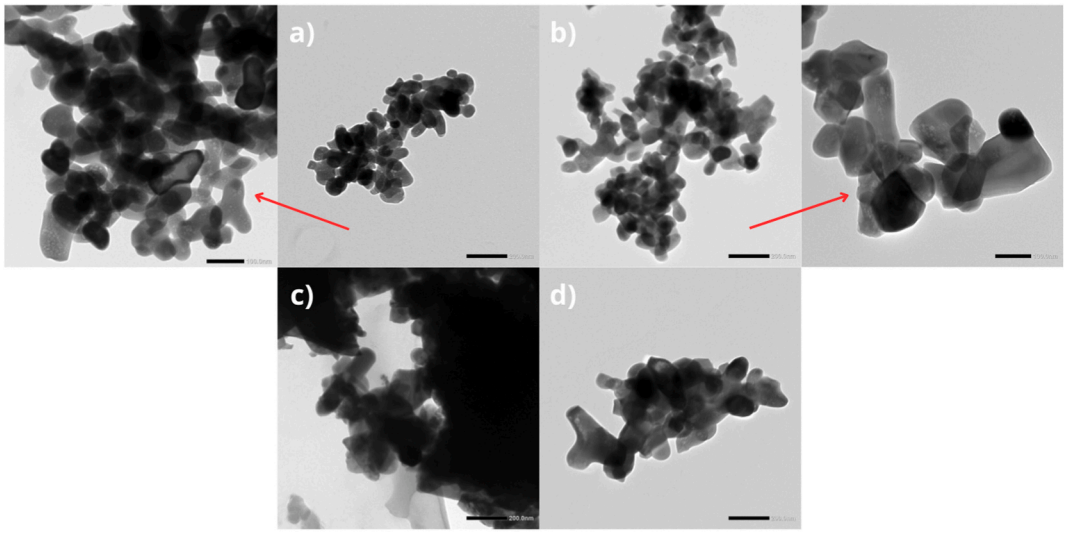

Fig. 4. TEM analyses of a) HApK800, b) HApK1000, c) HApA800, d) HApA1000. All main images were acquired with a scale bar of 200 nm, and the red arrows indicate higher magnification regions shown with a 100 nm scale bar to highlight porosity.

TEM observations showed that acidic synthesis promoted denser and more compact agglomerates compared to the porous, rod-like morphology obtained by the alkaline route.

Overall, the results confirm that the proposed route is a viable alternative for producing structurally stable hydroxyapatite and could provide a controlled platform to investigate the role of pH on phase stability and crystal growth, opening promising perspectives for tailoring bioceramic powders with controlled composition and crystallinity. The acidic medium, by enabling alternative ionic equilibria, may also facilitate targeted dopant incorporation and support the development of bioactive coatings and advanced biomedical ceramics. Future work will aim to optimize reaction parameters to suppress CaO formation and to assess the biological and mechanical performance of these materials under physiological conditions.

CRediT authorship contribution statement

Alícia C. Mattioni: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Júlia B. Cassel:** Writing – review & editing, Methodology, Formal analysis, Conceptualization. **Luiz Fernando Rodrigues Júnior:** Writing – review & editing, Supervision, Conceptualization. **Luis Alberto Loureiro dos Santos:** Writing – review & editing, Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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